Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Faces, Edges, and Vertices

In everyday life you meet many objects that are solid — a dice, a gift box, a brick. A solid shape that takes up space is called a three-dimensional (3-D) shape, and each 3-D shape can be described by three special properties:

A face is a flat surface of the shape.
An edge is a line where two faces meet.
A vertex (plural: vertices) is a corner, where edges meet.

Let's use these three words on two shapes you see all the time.

The cube — think of a dice. Every face of a cube is an identical square, all the same size. To count its parts:

It has 6 faces — the top, the bottom, and four around the sides.
It has 12 edges — each of the six square faces has four edges, but every edge is shared by two faces, so $6 \times 4 \div 2 = 12$.
It has 8 vertices — 4 on the top and 4 on the bottom.

The cuboid — think of a brick or a shoebox. Like a cube it has 6 faces, 12 edges and 8 vertices, but unlike a cube its faces are rectangles, and the faces come in three pairs of equal size: the two ends match, the two sides match, and the top and bottom match. A cube is actually a special cuboid where all six faces happen to be equal squares.

So cubes and cuboids share the same counts — 6 faces, 12 edges, 8 vertices — but a cube has all square faces while a cuboid has rectangular faces.</p>

Example 1 — Counting the cube
A dice is a cube. How many faces, edges and vertices does it have?

Solution:
Faces: 6 square faces (top, bottom, and four sides).
Edges: 12 (count 4 around the top, 4 around the bottom, and 4 joining them).
Vertices: 8 (4 on the top face, 4 on the bottom face).

Answer: 6 faces, 12 edges, 8 vertices.

Example 2 — Telling a cuboid from a cube
A shoebox is 30 cm long, 20 cm wide and 10 cm high. Is it a cube or a cuboid? How can you tell?

Solution:
A cube needs all six faces to be equal squares. This box has faces of lengths 30×20, 30×10, and 20×10, which are rectangles of different sizes, not equal squares. So it is a cuboid. It still has 6 faces, 12 edges and 8 vertices.

Practice

1. A sugar cube has the shape of a cube. Write down the number of faces, edges and vertices a cube has.
2. A box of cereal is a cuboid. Its top face is a rectangle measuring 8 cm by 3 cm. Explain why this box cannot be a cube.

3. A cuboid has 12 edges. Four of its faces are rectangles of size 5 cm by 3 cm, and the other two faces are squares. Which shape does this describe — a cube or a cuboid? Give a reason for your answer.

Section 2: Pyramids and Cones: Faces, Edges, and Vertices

A pyramid is a solid shape with a flat base and triangular faces that all meet at a single point called the apex. The shape of the base gives the pyramid its name. If the base is a square, it is a square pyramid (like the famous ones in Egypt); if the base is a triangle, it is a triangular pyramid.

Every pyramid shares these properties:
- It has one base, which can be any polygon (a triangle, a square, a pentagon, and so on).
- All the other faces are triangles, and they all meet at one point, the apex.
- It has edges (where two faces meet) and vertices (corners, where edges meet).

You can count the parts of a pyramid by looking at its base. A square-based pyramid has:
- a square base,
- 4 triangular faces (one for each side of the base),
- 1 apex (the point where the triangles meet),
- 5 faces in total (the base plus the 4 triangles),
- 8 edges (4 around the base and 4 sloping up to the apex),
- 5 vertices (4 at the corners of the base plus the apex).

A cone is different. A cone has a flat circular base and one curved surface that slopes up to a single point at the top, also called the apex or vertex. Because its base is a circle, a cone has no straight edges and no corners.

A cone has:
- 1 flat circular base,
- 1 curved surface (the sloping part),
- 1 vertex (the point at the top),
- 0 edges and 0 vertices on the base, because a circle has no corners.

So the big difference is this: a pyramid has straight edges and flat triangular faces, while a cone has a smooth curved surface and no edges at all.

Worked Example 1 — A triangular-based pyramid
A pyramid has a triangular base. How many faces, edges and vertices does it have?

Step 1: Count the faces. The base is 1 triangle, and one triangular face rises from each of its 3 sides. So faces = 1 base + 3 side triangles = 4 faces.

Step 2: Count the edges. There are 3 edges around the base and 3 edges sloping up to the apex. So edges = 3 + 3 = 6.

Step 3: Count the vertices. There are 3 corners on the base plus the apex at the top. So vertices = 3 + 1 = 4.

A triangular-based pyramid has 4 faces, 6 edges and 4 vertices.

Worked Example 2 — Comparing a square pyramid and a cone
A square-based pyramid and a cone stand side by side on a table.

For the square pyramid, count its faces, edges and vertices:
- Faces: 1 square base + 4 triangular sides = 5 faces.
- Edges: 4 base edges + 4 sloping edges = 8 edges.
- Vertices: 4 base corners + 1 apex = 5 vertices.

For the cone, count the same things:
- Faces: 1 flat circular base + 1 curved surface = 2 faces.
- Edges: none, because there is no straight edge where the circle meets the curved surface.
- Vertices: just 1, the point at the top.

So the pyramid has straight edges and flat triangular faces, while the cone has a curved surface and no edges. That is the key difference between them.

Practice

1. A pyramid has a square base. How many faces, edges and vertices does it have in total?
2. A cone has a flat circular base and one curved surface. How many faces, edges and vertices does a cone have?
3. Compare a triangular-based pyramid with a cone. Write down one thing about their edges that is different, and one thing they both have.
4. A pyramid's base has 5 sides (a pentagon). Without drawing it, work out how many faces, edges and vertices the pyramid has. Explain your counting.

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Faces, Edges, and Vertices

1. 6 faces, 12 edges, 8 vertices.
2. A cube must have all six faces as equal squares. A cereal box has rectangular faces of different sizes (for example its top is 8 cm by 3 cm, which is a rectangle, not a square), so it cannot be a cube.
3. A cuboid. A cube needs all six faces to be equal squares, but here four faces are rectangles and only two are squares, so it is a cuboid (a rectangular prism that is not a cube).

Pyramids and Cones: Faces, Edges, and Vertices

1. 5 faces, 8 edges, 5 vertices.

   Faces: the square base is 1 face and there are 4 triangular side faces (one from each side of the base), so 1 + 4 = 5 faces. Edges: 4 around the base plus 4 sloping up to the apex, so 4 + 4 = 8 edges. Vertices: 4 corners on the base plus 1 apex, so 4 + 1 = 5 vertices.
2. 2 faces, 0 edges, 1 vertex.

   A cone has 1 flat circular base and 1 curved surface, so 2 faces in total. Because the base is a circle, it has no corners, so it has 0 edges. It has 1 vertex at the point at the top.
3. The pyramid has 6 straight edges while the cone has 0 edges. Both have a flat base and both come to a single point (apex) at the top.

   The triangular pyramid has 6 straight edges (3 around the base and 3 sloping up), but a cone has no straight edges at all because its base is a circle. They are alike because both have a single flat base and both end in one point at the top (called the apex or vertex).
4. 6 faces, 10 edges, 6 vertices.

   Faces: 1 pentagon base + 5 triangular side faces = 6 faces. Edges: 5 around the base + 5 sloping up to the apex = 10 edges. Vertices: 5 corners on the base + 1 apex = 6 vertices.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-38ddd222	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____